Problem Chosen	2026 MCM/ICM Summary Sheet	Team Control Number
C		2626843

Counterfactual Analysis of Voting Rules Using Constrained Bayesian Inference and Rejection Sampling

Summary

Dancing with the Stars is a popular American TV series featuring dance competition among celebrities. Both fan votes and judges scores are taken into consideration to decide the final winner. How should the two different measurement be combined to achieve the optimal result? What factors can impact the contestants' performance, fan votes and judges score?

The dataset provided consists of information about the contestants in all 34 seasons and the judges scores they got in different weeks. Before modeling, we cleaned and normalized the given data to avoid the influence of missing data and the uncertainty of total votes.

First, we propose a **constrained Bayesian framework** to estimate unobservable weekly fan vote shares for DWTS Seasons 1–34. Modeling vote shares as a **truncated Dirichlet distribution**, we integrate weak signals (standardized judge scores, popularity inertia) and align with season-specific rules to match observed eliminations. We use **rejection sampling** to generate valid posterior samples, with consistency (Elimination Consistency Score, feasibility) and certainty (posterior entropy, 90% credible intervals) metrics as required.

Secondly, we leverage the previous fan vote estimation to conduct **counterfactual analysis of *DWTS*'s voting rules**. We test controversial cases (e.g., Jerry Rice, Bobby Bones), evaluate the Bottom-Two + judges' choice mechanism, and recommend a preferred method for future seasons, balancing fairness and fan engagement as requested.

Thirdly, we develop **mixed-effects models** to analyze how celebrity characteristics and pro dancers impact judges' scores and Model I-derived fan vote shares. We compare judges' and fans' responses to these factors, **propagate vote estimation uncertainty**, and link mechanisms to final performance via a **survival model**.

Next, we propose an **Adaptive Hybrid Rule** to combine judges' scores and fan votes. We used the risk indicators obtained in model I(**Posterior entropy, Outcome margin**) to assess the risk level of different weeks. For low risk weeks, we apply the original rank-based combination rule. For high risk weeks, we activate our stabilizer to find an optimal elimination choice.

Keywords: Constrained Bayesian Inference; Voting Rule Optimization; Adaptive Hybrid Rule; Mixed-Effects Model; Truncated Dirichlet Posterior; Rejection Sampling; Counterfactual Analysis

Contents

1 Introduction

1.1 Problem Background

Dancing with the Stars (DWTS), the U.S. adaptation of the "Strictly Come Dancing" franchise, has run for 34 seasons. It's weekly eliminations and final rankings depend on two core inputs: expert judges' technical scores and fan votes .

The show has adopted various vote-combining methodologies over time.[1] Seasons 1–2 used a rank-based method (sum of judge/fan ranks), but Season 2's controversy—Jerry Rice reaching the finals with five weeks of lowest judge scores—led to a shift to a percentage-based approach (sum of judge/fan vote percentages) for Seasons 3–27. Another controversy in Season 27 (Bobby Bones winning with consistently low judge scores) prompted Season 28 revisions: reverting to rank-based combination and having judges eliminate one of the bottom two couples identified by combined votes.

While data on judge scores, celebrity traits (age, industry), partner info, and competition results are available, fan vote totals remain undisclosed. This gap drives the need for modeling to estimate fan votes, compare the two combination methods, analyze controversial outcomes, and propose an improved voting system—with insights to balance fairness, technical integrity, and audience engagement.[2][3]

Figure 1: Dancing with the Stars

1.2 Literature Review

Relevant literature aligns with three core domains of this study: talent show mechanism analysis, voting rule optimization, and statistical modeling for latent variables.

Talent Show Mechanisms and Audience Behavior

Bonner [2] analyzed *Strictly Come Dancing* (DWTS prototype), identifying that celebrity traits shape audience support beyond technical performance—informing Task 3's analysis of celebrity/pro partner impacts. Wood [3] noted fan voting depends on both performance and emotional connection, supporting Model I's integration of "popularity inertia" as a prior signal. Mawhinney et al. [4] drew parallels between talent show ranking and prognostic models, reinforcing Model I's "rule-consistent inference" rigor.

Voting Rule Fairness

Balinski and Laraki [5] established a framework for evaluating rank/score-based voting rules, showing rank-based methods reduce volatility—directly supporting Task 2's rule comparison and our use of "flip probability" to quantify outcome divergence.

Statistical Methods

For Model I's rejection sampling, Flury [6] simplified acceptance-rejection sampling for truncated distributions (e.g., truncated Dirichlet posterior for fan votes). For Model II's mixed-effects models, Bates et al. [7] provided practical implementation for hierarchical data (repeated weekly observations), addressing our data structure. Papke and Wooldridge [8] pioneered fractional logit models for proportional data (e.g., fan vote shares), underpinning Model II's stabilized logit transform for fan vote regression.

1.3 Restatement of the Problem

We need to analyze the provided data about Dancing with the Stars(DWTS) and complete the following tasks.

Task 1:

1. Develop a mathematical model to estimate fan votes, then measure if the estimated fan votes are consistent with who was eliminated each week.
2. Evaluate the certainty of the produced fan vote totals, then consider if that certainty is the same for each contestant/week.

Task 2:

1. Apply the two approaches used by the show to combine judge and fan votes across seasons to each seasons. If different results were produced, then consider if one method favors fan votes more than the other.
2. Consider the examples where the judges scores and fan votes disagree with each other. Can the two different methods lead to different results, then discuss if including the additional approach of having judges choose which of the bottom two couples to eliminate each week makes any difference.
3. Select an optimal method between the two methods, then discuss if the additional approach of judges choosing from the two bottom couples should be adopted.

Task 3:

Develop a model to analyze the impact of various pro dancers and the characteristics for the celebrities included in the data. Evaluate how much do these factors impact a celebrity's performance, then compare the impact of these factors on judges scores and fan votes.

Task 4:

Develop a new system to measure the performance of celebrities with judges scores and fan votes, then explain why the new system is better than the previous systems.

1.4 Our Work

Our work flow of this paper is shown in Figure 2 on Page 5.

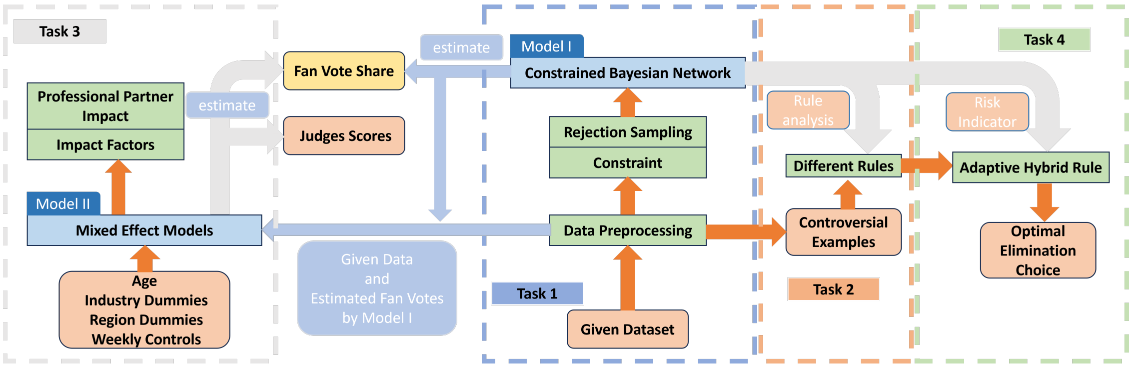

Figure 2: Workflow of our research.

2 Assumptions and Notations

To ensure model identifiability and consistency with the official voting mechanics, we introduce the following assumptions. These assumptions do not impose behavioral restrictions on individual voters, but rather reflect structural constraints implied by the show format and data availability.

Assumption 1: In each season, the fan voting outcome of a given week depends only on the contestants who are still active in that week.

Justification 1: Once a contestant is eliminated, the contestant no longer participates in the competition. Therefore, fan votes in a given week are allocated exclusively among the remaining contestants, and eliminated contestants do not receive any votes thereafter.

Assumption 2: The total number of fan votes in each week is unknown but fixed, and only the relative proportions of votes among contestants influence elimination outcomes.

Justification 2: The exact fan vote counts are not publicly available. Both the ranking-based and percentage-based elimination rules depend only on the relative ordering or percentage share of votes. Consequently, modeling fan votes as normalized proportions is sufficient and avoids introducing unidentifiable parameters.

Assumption 3: Any changes in voting or elimination mechanisms occur only across different seasons.

Justification 3: Different seasons of the show adopt different vote-combination rules (e.g., rank-based versus percentage-based), and that later seasons introduce additional mechanisms such as the *Bottom Two* followed by judges' decisions. There is no evidence of mid-season rule changes; thus, it is reasonable to assume that the elimination rule remains fixed within each season. If minor mid-season deviations did occur, they would primarily manifest as weeks with low elimination consistency scores, which are explicitly identified and treated as high-uncertainty cases in subsequent analysis.

Notations

The main symbols used throughout this paper are summarized as follows in Table 1. Other infrequently-used symbols will be introduced once they are used.

Table 1: Notation for DWTS Voting & Scoring System

Notation	**Definition**
s	Season index.
w	Week index within a season.
i	Index of a contestant.
$J_{i,w}$	Total judge score received by contestant i in week w.
$v_{i,w}$	Proportion of fan votes received by contestant i in week w.
$A_{s,w}$	Set of contestants active in season s during week w.
$E_{s,w}$	Set of contestants eliminated at the end of week w in season s.

3 Data Preprocessing

The raw dataset contains contestant attributes, season outcomes, professional partner information, and weekly judges' scores recorded in a wide format as `weekX_judgeY_score`. Because seasons vary in length and the number of judges differs across seasons, the raw data must be transformed into consistent season–week panels before modeling. Our preprocessing produces two analysis-ready datasets: (i) a contestant–week long table used in Tasks 1–3, and (ii) a contestant-level summary table (survival-style) used in Task 3 outcome modeling.

Table 2 summarizes the key preprocessing steps.

Step	**Description**
Long-table construction	Reshape `weekX_judgeY_score` columns into a contestant–week long table (one row per contestant-week).
Judges' score aggregation	Sum all weekly judges' scores to obtain $J_{i,w}$; treat missing values as zero.
Missing value handling	Replace `N/A` with 0 to account for absent judges or non-existent weeks.
Valid week identification	Retain season-weeks with at least one positive weekly total score.
Active contestant detection	Define active contestants as those with $J_{i,w} > 0$.
Elimination inference	Infer eliminations by comparing active sets of consecutive valid weeks.
Week-level covariates	Compute remaining contestants $n_{s,w} = \lvert A_{s,w}\rvert$ and eliminations $k_{s,w} = \lvert E_{s,w}\rvert$ for each season-week.
Judge-score standardization	Standardize weekly judges' totals among active contestants to obtain $z_{i,w}$ used in Task 3 regressions.
Fan-vote integration (Task 3)	Merge inferred fan vote posterior mean and uncertainty (e.g., $\hat{v}_{i,w}$ and $\mathrm{sd}(v_{i,w})$) into the long table for downstream modeling.

Table 2: Summary of data preprocessing procedures.

Note on missing values. Replacing missing judges' scores with zero reflects the absence of a recorded score (e.g., due to a reduced judging panel or non-performance weeks), rather than an imputation of poor performance. Season-weeks without any positive total score are subsequently filtered during the valid-week identification step, ensuring that such placeholders do not distort downstream inference.

4 Model I: Estimating Fan Votes for each Contestant

In this part we develop a model to estimate fan votes for each contestant-week. Then we measure the consistency and certainty of our estimation model.

4.1 Modeling Intuition

We characterize *all possible fan vote distributions* that could have produced the observed elimination outcomes under the official show rules. Among these feasible distributions, we then quantify which vote shares are more plausible and how uncertain those estimates are. This perspective allows uncertainty to be treated as an intrinsic feature of the model, rather than as an afterthought.

Formally, we address both objectives by inferring a *posterior distribution* of fan vote shares constrained by the show mechanics.

Because absolute fan vote totals are not available and are not identifiable from the dataset, we model fan votes as vote *shares*. For each season-week (s, w) with active set $A_{s,w}$ of size n, we define the latent vote-share vector

$$\mathbf{v}_{s,w} = (v_{1,w}, \ldots, v_{n,w}), \quad v_{i,w} \geq 0, \quad \sum_{i=1}^{n} v_{i,w} = 1.$$

If a vote-count scale is needed for interpretation, we report pseudo-votes $\tilde{V}_{i,w} = N_0 v_{i,w}$ under a fixed constant N_0; all comparisons and eliminations depend only on $\mathbf{v}_{s,w}$.

4.2 Prior: Dirichlet Vote Shares with Weak Structure

A Dirichlet distribution is a natural prior on the simplex:

$$\mathbf{v}_{s,w} \sim \text{Dirichlet}(\boldsymbol{\alpha}_{s,w}), \qquad \alpha_{i,w} = \kappa\, \mu_{i,w},$$

where $\kappa > 0$ controls concentration and $\boldsymbol{\mu}_{s,w}$ is a simplex-valued prior mean.

To avoid an overly diffuse posterior (which can occur if the prior is fully uniform), we incorporate two weak and interpretable signals into $\mu_{i,w}$:

- **Performance signal** via judges' scores;
- **Popularity inertia** via last week's inferred vote shares.

Interpretation of the prior. Importantly, judges' scores are used only to construct a *weak prior belief* about relative fan support, not as direct observations of fan voting behavior. All final inferences are driven by consistency with observed elimination outcomes under the show rules, rather than by regression on judges' scores.

Let $J_{i,w}$ be weekly total judges' scores and define the within-week standardized score

$$z_{i,w} = \frac{J_{i,w} - \overline{J}_w}{\text{sd}(J_{\cdot,w}) + \epsilon}.$$

Let $\widehat{v}_{i,w-1}$ denote last week's posterior mean share (restricted to contestants still active in week w); for the first week, set $\widehat{v}_{i,0} = 1/n$. We define

$$\mu_{i,w} \propto \exp\big(\beta z_{i,w} + \rho \log(\widehat{v}_{i,w-1} + \epsilon)\big), \qquad \sum_i \mu_{i,w} = 1,$$

where $\beta \geq 0$ controls the performance-to-popularity linkage and $\rho \geq 0$ controls popularity inertia.

4.3 Show Mechanics as Constraints (Season-Dependent)

The problem statement describes two historical vote-combination schemes and a later *bottom-two then judges' decision* mechanism. We encode these mechanics to define, for each week, a feasible set $C_{s,w}$ of vote-share vectors that could have produced the observed outcome.

Percentage-Based Combination (Seasons 3–27)

Define the judges' score share

$$u_{i,w} = \frac{J_{i,w}}{\sum_{j \in A_{s,w}} J_{j,w}},$$

and the combined score

$$c_{i,w} = u_{i,w} + v_{i,w}.$$

For single elimination with $E_{s,w} = \{e\}$, feasibility requires

$$c_{e,w} \le c_{i,w} \quad \forall i \ne e.$$

For multiple eliminations with $|E_{s,w}| = k$, we require every eliminated contestant to lie among the k smallest combined scores.

Rank-Based Combination (Seasons 1–2 and assumed Seasons 28–34)

Let $r^J_{i,w}$ be the ordinal rank of $J_{i,w}$ (rank 1 is best), and $r^V_{i,w}$ be the ordinal rank of $v_{i,w}$ (rank 1 is largest vote share). The combined rank is

$$R_{i,w} = r^J_{i,w} + r^V_{i,w}.$$

The eliminated contestant is the one with the *largest* $R_{i,w}$; for multiple eliminations we require the eliminated set to lie among the worst k combined ranks.

Ties in judges' scores can occur in the data. When computing ordinal ranks $r^J_{i,w}$, we assign *mid-ranks* (average ranks) to tied contestants. For fan vote shares drawn from a continuous Dirichlet distribution, ties in $v_{i,w}$ occur with probability zero and can be ignored.

Bottom-Two + Judges' Choice (assumed Seasons 28–34)

When the show determines a bottom two and then judges select one of those two to eliminate, it is generally incorrect to enforce that the eliminated contestant must be uniquely worst under the combined metric. Instead, we impose the weaker constraint

$$E_{s,w} = \{e\} \implies e \in \text{Bottom2}(\cdot),$$

where $\text{Bottom2}(\cdot)$ is computed from the combined score or rank. This produces a larger feasible set and appropriately increases posterior uncertainty.

Finals Week Ordering

In the finals, combined judges' and fan scores are used to rank the remaining contestants. Accordingly, for the final competition week we impose pairwise ordering constraints consistent with the observed final placements, using the season-specific combination scheme.

4.4 Posterior: Truncated Dirichlet Distribution

Given the prior and feasibility constraints, the posterior distribution of fan vote shares is

$$p(\mathbf{v}_{s,w} \mid \text{observations}) \propto \text{Dirichlet}(\mathbf{v}_{s,w}; \boldsymbol{\alpha}_{s,w})\, \mathbb{I}(\mathbf{v}_{s,w} \in C_{s,w}).$$

This defines a *truncated Dirichlet distribution* supported on a rule-induced subset of the simplex.

4.5 Inference via Rejection Sampling

We approximate the truncated posterior using acceptance–rejection sampling:

1. Draw $\mathbf{v}^{(m)} \sim \text{Dirichlet}(\boldsymbol{\alpha}_{s,w})$, $m = 1, \dots, M$;
2. Apply the season-specific rule to $\mathbf{v}^{(m)}$ and compute the implied elimination set (or bottom-two set);
3. Accept $\mathbf{v}^{(m)}$ if and only if it satisfies the regime-specific feasibility constraint defined in Section 4.3 (e.g., exact eliminated set under percent/rank rules, or membership in the Bottom-Two set under the Bottom-Two + judges' choice regime, and finals ordering constraints when applicable).

By classical results on acceptance–rejection sampling, the accepted samples are distributed according to the target truncated distribution (see, e.g., [6]). This approach is robust to non-smooth, discontinuous, or non-convex feasible regions, which naturally arise under ranking-based rules.

Rule-consistent representative estimate.

Under rank-based constraints, the posterior mean can fall outside the feasible region. To provide a single representative estimate that is *guaranteed* to reproduce the observed elimination, we select a rule-consistent representative among accepted draws:

$$\widetilde{\mathbf{v}}_{s,w} = \arg\max_{\mathbf{v}^{(m)} \in C_{s,w}} \log \text{Dirichlet}(\mathbf{v}^{(m)}; \boldsymbol{\alpha}_{s,w}).$$

We report $\widetilde{\mathbf{v}}_{s,w}$ as a consistent point estimate and use the full accepted sample set to quantify uncertainty.

Posterior draws retained for downstream analysis.

We retain the accepted draws $\{\mathbf{v}^{(m)}\}_{m=1}^{N_{\text{acc}}}$ for each (s, w). These draws will be re-used in Section 4.8 to evaluate alternative voting rules under posterior uncertainty.

4.6 Certainty and Consistency Measures

Certainty(Uncertainty) Measures

For each contestant-week we compute:

- posterior mean and standard deviation of $v_{i,w}$;
- a 90% credible interval $[q_{0.05}(v_{i,w}),\ q_{0.95}(v_{i,w})]$;
- a scale-free uncertainty measure $\text{sd}(v_{i,w})/(\mathbb{E}[v_{i,w}] + \epsilon)$.

To characterize week-level uncertainty, we compute:

- **Outcome margin:** a week-level "knife-edge" indicator defined as the gap between the worst and second-worst contestants under a chosen rule operator. Concretely, under the default rank-based combination we compute each contestant's combined rank $R_{i,w} = r^J_{i,w} + r^V_{i,w}$ and define
$$M_{s,w} = R_{(n-1),w} - R_{(n),w},$$
where $R_{(n),w}$ and $R_{(n-1),w}$ are the largest and second-largest combined ranks among active contestants in week (s, w) (larger is worse). Smaller $M_{s,w}$ indicates a more knife-edge elimination.

- **Support concentration:** entropy of the posterior mean vote shares
$$H(\widehat{\mathbf{v}}_{s,w}) = -\sum_i \widehat{v}_{i,w} \log(\widehat{v}_{i,w} + \epsilon).$$

Low entropy and large outcome margins indicate stable fan preferences, whereas high entropy and narrow margins suggest volatile or polarized audience behavior.

Visualization of posterior uncertainty.

To make the uncertainty in inferred fan support tangible, we visualize the posterior vote-share distributions for a representative season-week. In particular, Figure 3 shows how strongly (or weakly) the elimination is supported by fan voting once the show rules are enforced.

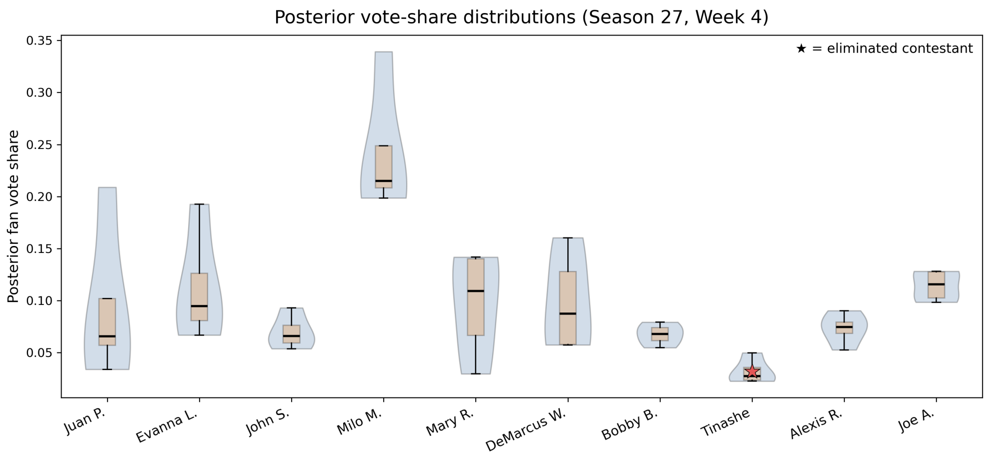

Figure 3: Posterior vote-share distributions (violin/box) for a representative season-week. The eliminated contestant is highlighted. Wide distributions and overlap indicate higher uncertainty.

Preview for Task 2 (judge–fan disagreement).

Preview for rule-level analysis.

In Section 4.8, we examine the tension between technical scores and inferred popularity by plotting judges' score shares against posterior mean fan vote shares, highlighting controversial contestants.

A bridge to rule counterfactuals.

For an overview across season-weeks, we summarize three quantities at the week level: (i) the elimination consistency score $\mathrm{ECS}_{s,w}$, (ii) the eliminated contestant's judges' score share $u_{e,w}$, and (iii) the entropy of the posterior mean fan vote shares $H(\widehat{\mathbf{v}}_{s,w})$. These diagnostics help interpret when alternative rule choices are likely to produce different outcomes.

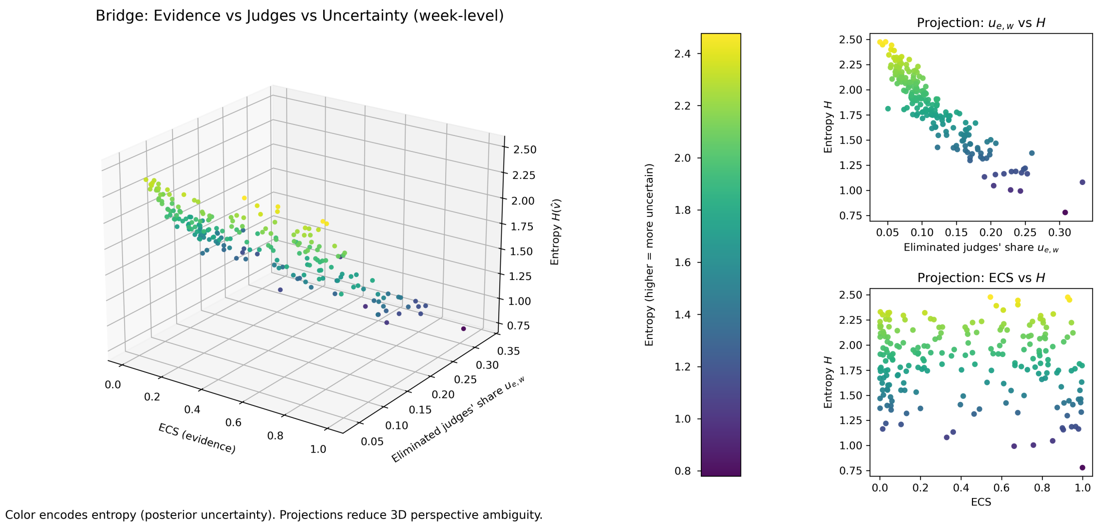

Figure 4: Bridge visualization across season-weeks: elimination consistency $\mathrm{ECS}_{s,w}$, eliminated judges' share $u_{e,w}$, and posterior entropy $H(\widehat{\mathbf{v}}_{s,w})$. Two projections are included to avoid 3D perspective ambiguity.

Consistency Measures

We quantify consistency using three complementary diagnostics:

- **Feasibility:** whether at least one accepted draw exists for a given week.
- **Elimination Consistency Score (ECS):**

$$\mathrm{ECS}_{s,w} = \frac{N_{\mathrm{acc}}}{M},$$

  where N_{acc} is the number of accepted samples. This score represents the probability that the observed elimination is compatible with the model assumptions and the show mechanics.

- **Representative consistency:** applying the official rule to $\widetilde{\mathbf{v}}_{s,w}$ reproduces the observed elimination (guaranteed by construction).

We visualize $\mathrm{ECS}_{s,w}$ across weeks to identify (i) highly stable eliminations with large model evidence and (ii) low-evidence weeks where unobserved popularity shocks are likely.

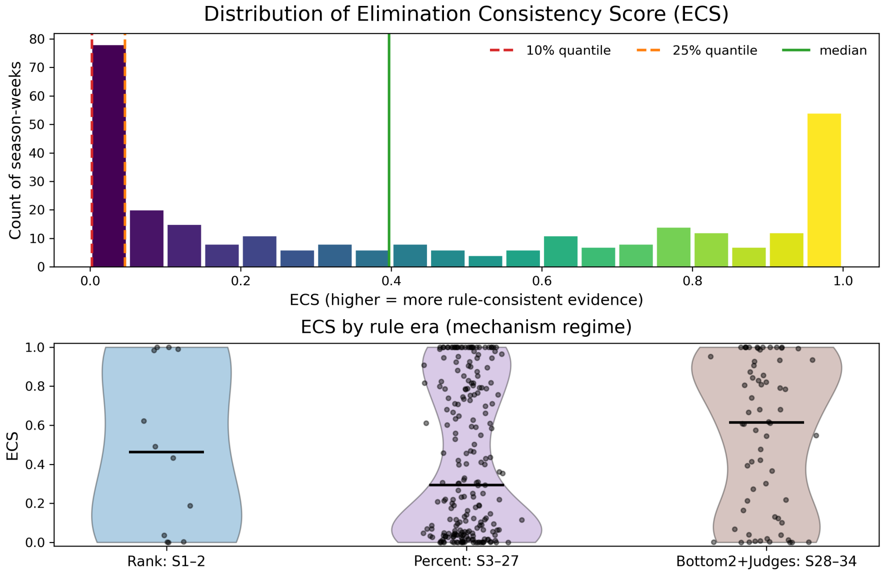

Figure 5: Elimination Consistency Score (ECS) across season-weeks. Higher values indicate stronger consistency between inferred fan votes and observed eliminations under the assumed show mechanics.

4.7 Hyperparameter Calibration (Recommended)

The hyperparameters (β, ρ, κ) control how strongly judges' scores and popularity inertia influence the prior, as well as how concentrated the prior distribution is. Any fixed choice yields a valid model; however, to avoid arbitrary tuning, we adopt a simple data-informed calibration.

Specifically, we define a season-level compatibility score

$$\mathrm{Score}(s) = \sum_{w \in W_s} \log(\mathrm{ECS}_{s,w} + \epsilon),$$

where $\mathrm{ECS}_{s,w}$ is the elimination consistency score for week w and W_s is the set of valid weeks in season s.

We select (β, ρ, κ) by a coarse grid search that maximizes this score and then verify that the main qualitative conclusions remain stable for nearby parameter values (sensitivity analysis). This step improves robustness but is not required for model validity.

4.8 Rule-Based Counterfactual Analysis

Task 2 asks how outcomes would differ under alternative official vote-combination rules. Because the true fan votes are unobserved, we perform a *posterior counterfactual* analysis: for each season-week (s, w), we treat posterior draws $\mathbf{v}_{s,w} \sim p(\mathbf{v}_{s,w} \mid \text{data})$ as plausible latent fan preferences, and re-apply different voting rules to the same $(\mathbf{J}_{s,w}, \mathbf{v}_{s,w})$.

Rule operators

Let $\mathcal{R} \in \{\text{pct}, \text{rank}\}$ denote the combination rule. Given judges scores $\mathbf{J}_{s,w}$ and a vote-share vector $\mathbf{v}_{s,w}$, define the rule-implied elimination set (for a k-elimination week) as

$$\mathcal{E}^{\mathcal{R}}_{s,w}(\mathbf{J}, \mathbf{v}; k).$$

For the percentage rule, contestants are ordered by $c_{i,w} = u_{i,w} + v_{i,w}$ (smaller is worse). For the rank rule, contestants are ordered by combined rank $R_{i,w} = r^J_{i,w} + r^V_{i,w}$ (larger is worse).

We also define the Bottom-Two set under rule $\mathcal{R}$ as

$$\mathcal{B}^{\mathcal{R}}_{s,w}(\mathbf{J}, \mathbf{v}) = \text{two worst contestants under rule } \mathcal{R}.$$

To emulate the *judges' choice* step, we consider a conservative proxy: judges eliminate the contestant with the smaller weekly judges score among the bottom two,

$$\mathcal{E}^{\mathcal{R},\text{JC}}_{s,w}(\mathbf{J}, \mathbf{v}) = \arg \min_{i \in \mathcal{B}^{\mathcal{R}}_{s,w}(\mathbf{J},\mathbf{v})} J_{i,w}.$$

This proxy is transparent and captures the intended effect of increasing the influence of judges once the bottom two are determined.

Posterior counterfactual metrics

Let $\{\mathbf{v}^{(m)}_{s,w}\}_{m=1}^{N_{\text{acc}}}$ be accepted posterior draws for (s, w). We quantify the probability that two rules disagree on who leaves:

$$p^{\text{flip}}_{s,w} = \frac{1}{N_{\text{acc}}} \sum_{m=1}^{N_{\text{acc}}} \mathbb{I}\Big(\mathcal{E}^{\text{pct}}_{s,w}(\mathbf{J}, \mathbf{v}^{(m)}; k) \neq \mathcal{E}^{\text{rank}}_{s,w}(\mathbf{J}, \mathbf{v}^{(m)}; k)\Big).$$

When $k > 1$ (multiple eliminations), we additionally report the Jaccard similarity

$$\text{Jac}_{s,w} = \frac{\left|\mathcal{E}^{\text{pct}}_{s,w} \cap \mathcal{E}^{\text{rank}}_{s,w}\right|}{\left|\mathcal{E}^{\text{pct}}_{s,w} \cup \mathcal{E}^{\text{rank}}_{s,w}\right|},$$

computed for each draw and averaged over posterior samples.

To measure whether a rule tends to empower fans over judges, we compare the eliminated contestants' fan-rank and judge-rank. For a k-elimination week, define

$$\Delta^{\mathcal{R}}_{s,w} = \frac{1}{k} \sum_{e \in \mathcal{E}^{\mathcal{R}}_{s,w}} (r^V_{e,w} - r^J_{e,w}),$$

and report its posterior mean. Larger $\Delta^{\mathcal{R}}_{s,w}$ indicates that contestants who are relatively *more favored by judges than by fans* are more likely to be eliminated, suggesting stronger fan influence under that rule.

Cross-season comparison: rank vs percent

We aggregate the week-level flip probabilities across each season:

$$\mathrm{EFR}(s) = \frac{1}{|W_s|} \sum_{w \in W_s} p_{s,w}^{\mathrm{flip}},$$

and compare the distribution of $\mathrm{EFR}(s)$ across seasons. High $\mathrm{EFR}(s)$ indicates that rule choice materially changes outcomes in that season, while low values suggest that the season is robust to rule design.

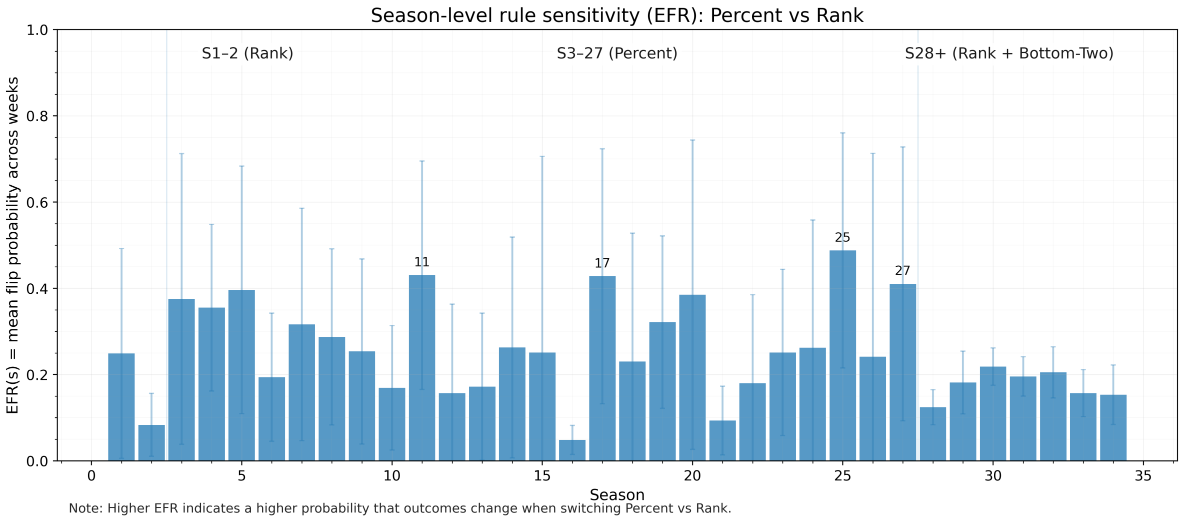

Figure 6: Season-level rule disagreement $\mathrm{EFR}(s)$ under posterior counterfactual analysis.

We also compare the season-averaged fan-influence index

$$\overline{\Delta}^{\mathcal{R}}(s) = \frac{1}{|W_s|} \sum_{w \in W_s} \mathbb{E}[\Delta_{s,w}^{\mathcal{R}}],$$

to assess whether one method systematically amplifies fan votes relative to judges.

Controversial contestants as stress tests

For each controversial celebrity (e.g., Jerry Rice in Season 2; Bobby Bones in Season 27), we treat their trajectory as a stress test for the voting rules. Using posterior draws, we compute the probability that the contestant would be eliminated in each week under each rule and under the Bottom-Two + judges' choice proxy. In finals weeks, we compute the posterior probability of each final placement under each rule.

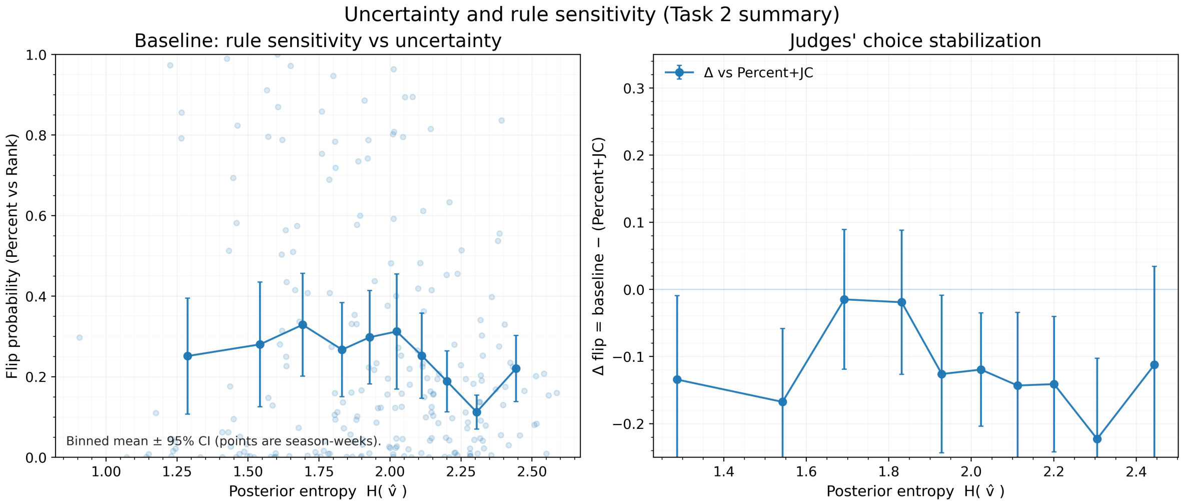

Figure 7: Uncertainty, Rule Sensitivity, & Bottom-Two Stabilization
Left: Higher fan vote entropy (uncertainty) → greater rule sensitivity.
Right: Bottom-Two mechanism reduces rule-driven outcome disagreement, especially in high-entropy weeks.

Impact of the Bottom-Two + judges' choice mechanism

Finally, we evaluate how the Bottom-Two mechanism alters outcomes relative to pure percent/rank rules. Using the proxy $\mathcal{E}^{\mathcal{R},\mathrm{JC}}_{s,w}$, we compute additional flip rates between (i) percent vs percent+JC and (ii) rank vs rank+JC. We interpret reductions in flip probability as increased outcome stability, particularly in high-entropy weeks where fan support is diffuse.

4.9 Implications for Rule Design and Recommendation

Our counterfactual analysis treats posterior fan vote shares as a proxy of latent audience preference and re-applies alternative official rules to the same season-week inputs $(\mathbf{J}_{s,w}, \mathbf{v}_{s,w})$. Across seasons, the rule-disagreement diagnostics (flip probability $p^{\text{flip}}_{s,w}$ and season-level $\mathrm{EFR}(s)$), together with the fan–judge tension measures (e.g., $\Delta^{\mathcal{R}}_{s,w}$ and the judge-share vs fan-share stress tests), reveal two consistent patterns.

Percent vs rank: when do outcomes change?

Weeks that exhibit diffuse or volatile fan support (high posterior entropy $H(\widehat{\mathbf{v}}_{s,w})$ and small outcome margins) are precisely the weeks in which the choice of combination rule is most consequential. In such "knife-edge" weeks, the percentage rule is more sensitive to large vote-share swings because it aggregates on the original scale (sum of shares), whereas the rank rule compresses magnitude differences into ordinal information. Consequently, the percent rule tends to amplify extreme popularity advantages, while the rank rule yields more stable eliminations when fan support is broadly distributed.

Fairness interpretation: balancing technical merit and audience engagement.

From the producers' perspective, a voting system should preserve *technical integrity* (so that strong performances are not systematically punished) while still respecting *audience engagement* (so that fan participation meaningfully matters). Our posterior bias summaries indicate that the rank-based scheme provides a more consistent trade-off: it prevents a single popularity surge from dominating the combined outcome, yet still allows sustained fan support to influence placements. In contrast, the percentage-based scheme is more likely to allow large fan-vote imbalances to overturn technical rankings, which aligns with the mechanism behind several controversy-style outcomes.

Recommendation 1: adopt a rank-based combination rule as the default.

Based on the above evidence, we recommend using the **rank-based** combination as the default vote-combination rule for future seasons. This choice is supported by (i) lower rule-disagreement in most weeks (smaller $\mathrm{EFR}(s)$ when compared counterfactually), and (ii) reduced sensitivity to vote-share magnitude shocks in high-entropy weeks, improving outcome stability without eliminating the role of fan participation.

Recommendation 2: retain the Bottom-Two + judges' choice mechanism as a stabilizer.

We further recommend retaining the **Bottom-Two + judges' choice** mechanism as a targeted stabilizer, particularly in weeks identified as high-uncertainty regimes (large $H(\widehat{\mathbf{v}}_{s,w})$ and small margins). Under posterior counterfactual simulation, this mechanism reduces volatility by constraining the final elimination decision to a small set of at-risk couples and re-introducing a technical tie-breaker. Importantly, it does *not* remove fan influence—fans still determine the Bottom-Two—but it mitigates the probability that a single extreme popularity shock alone decides the exit in otherwise ambiguous weeks.

Practical implementation note.

The above findings suggest that stabilization is most valuable only in a subset of "high-risk" weeks (high entropy and/or narrow margins). In Task 4, we formalize this idea into a transparent trigger-based voting system and evaluate it under the same posterior counterfactual framework.

5 Model II: Mechanism Analysis and Outcome Impact

5.1 Goals and Connection to Model I

This section extends the fan-vote inference results of Model I to explain *why* judges' scores and fan support may diverge, and *how much* professional partners and celebrity characteristics affect competition performance. The key outputs of Model II are: (i) partner/celebrity effects on judges' weekly scores, (ii) partner/celebrity effects on inferred fan vote shares, (iii) whether judges and fans respond to the same factors in the same way, and (iv) how these mechanisms translate into "how well" a celebrity ultimately does (e.g., surviving longer and finishing higher).

A central modeling issue is dependence: observations are repeated across weeks for the same celebrity, and the same professional partner appears with multiple celebrities. Therefore, we use hierarchical (mixed-effects) models with partial pooling rather than naive OLS.

5.2 Data for Mechanism Modeling

We use the contestant–week long table constructed in the data pipeline: each row corresponds to an active contestant-week (s, w, i) and includes (i) weekly judges scores (standardized within season-week), (ii) week/season indices and week-level controls (e.g., number of remaining contestants), (iii) professional partner identifiers, and (iv) celebrity covariates (age, industry, region, etc.).

Fan vote share $v_{swi} \in (0, 1)$ is not directly observed and is provided by Model I as posterior information (e.g., posterior mean/SD or posterior draws). In Model II we explicitly incorporate this uncertainty when fitting the fan mechanism model.

5.3 Mechanism Models: Judges vs Fans

Judges' Scores: Linear Mixed-Effects Model

Let $z_{swi}^{(J)}$ be the standardized judges score for contestant i in season s, week w. We fit the mixed-effects model:

$$z_{swi}^{(J)} = \alpha_J + \mathbf{x}_{swi}^\top \boldsymbol{\beta}_J + u_{\text{pro}(i)}^{(J)} + u_{\text{celeb}(i)}^{(J)} + \delta_s + \gamma_w + \varepsilon_{swi}^{(J)}, \tag{1}$$

where $\mathbf{x}_{swi}$ contains celebrity covariates (age, industry dummies, region dummies) and week-level controls (e.g., remaining contestants $n_{s,w}$). We include season fixed effects δ_s and week fixed effects γ_w to control for season-level scaling/rule differences and week-level difficulty trends.

Random intercepts $u_{\text{pro}(i)}^{(J)} \sim \mathcal{N}(0, \sigma_{\text{pro},J}^2)$ capture professional partner impact with partial pooling, and $u_{\text{celeb}(i)}^{(J)} \sim \mathcal{N}(0, \sigma_{\text{celeb},J}^2)$ absorb persistent unobserved celebrity-level heterogeneity. Compared to OLS, this structure prevents overstating significance from repeated measures and yields stable partner-effect estimates even for partners with limited data.

Fan Vote Share: Stabilized Logit + Mixed Effects

Fan support is modeled on the vote-share scale $v_{swi} \in (0, 1)$ inferred by Model I. Because v_{swi} is bounded and typically heteroskedastic, we avoid direct linear regression on v_{swi}. We use a stabilized logit transform:

$$y_{swi}^{(V)} = \log\left(\frac{v_{swi} + \epsilon}{1 - v_{swi} + \epsilon}\right), \qquad \epsilon \in (0, 0.01), \tag{2}$$

and fit the parallel mixed-effects model:

$$y_{swi}^{(V)} = \alpha_V + \mathbf{x}_{swi}^\top \boldsymbol{\beta}_V + u_{\text{pro}(i)}^{(V)} + u_{\text{celeb}(i)}^{(V)} + \delta_s + \gamma_w + \varepsilon_{swi}^{(V)}. \tag{3}$$

This makes the fan model comparable to the judge model under the same dependence structure, and keeps predictions on an unconstrained scale.

Do Judges and Fans React the Same Way?

Task requirements emphasize whether the same factors affect judges and fans similarly. Thus, we compare coefficients by contrasts:

$$\Delta\beta_k = \beta_{V,k} - \beta_{J,k}, \tag{4}$$

where covariates in $\mathbf{x}_{swi}$ are standardized so magnitudes are comparable. To respect dependence, we obtain uncertainty for $\Delta\beta_k$ using a *cluster bootstrap* (resampling celebrities and keeping all their weeks), and report confidence intervals for each contrast. If the interval for $\Delta\beta_k$ excludes 0, we conclude that factor k operates differently for judges versus fans. The results are shown in Figure 9.

5.4 Uncertainty Propagation from Model I

A key advantage of our approach is that fan votes are inferred with uncertainty. Using only a posterior mean would understate downstream uncertainty. Therefore, we propagate Model I uncertainty into the fan mechanism model via posterior resampling:

Posterior resampling. For $m = 1, \ldots, M$, draw $v_{swi}^{(m)}$ from the Model I posterior (or an approximate distribution parameterized by posterior mean and SD when only summaries are retained),

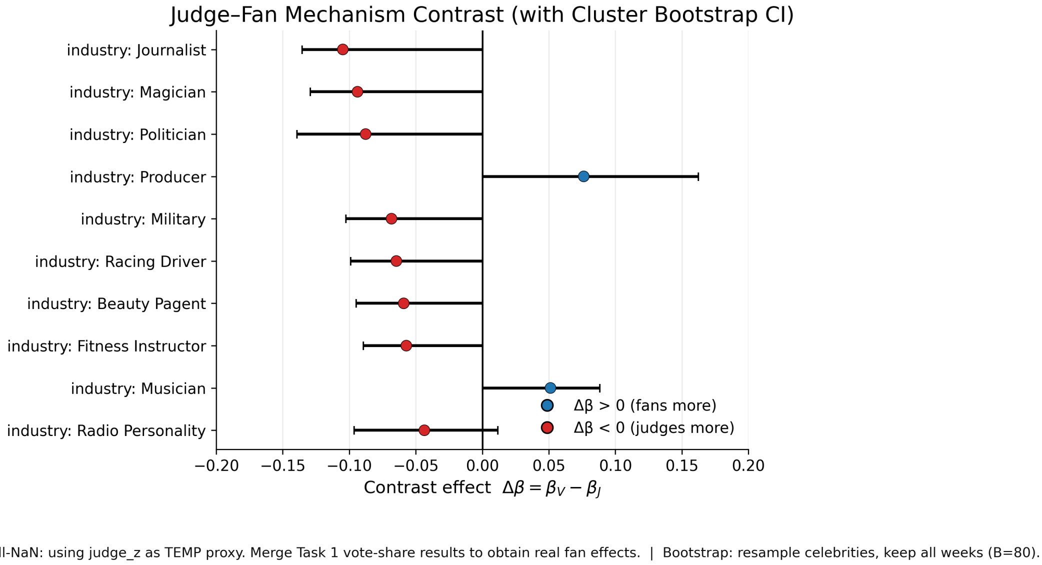

Figure 8: The results of cluster bootstrap.

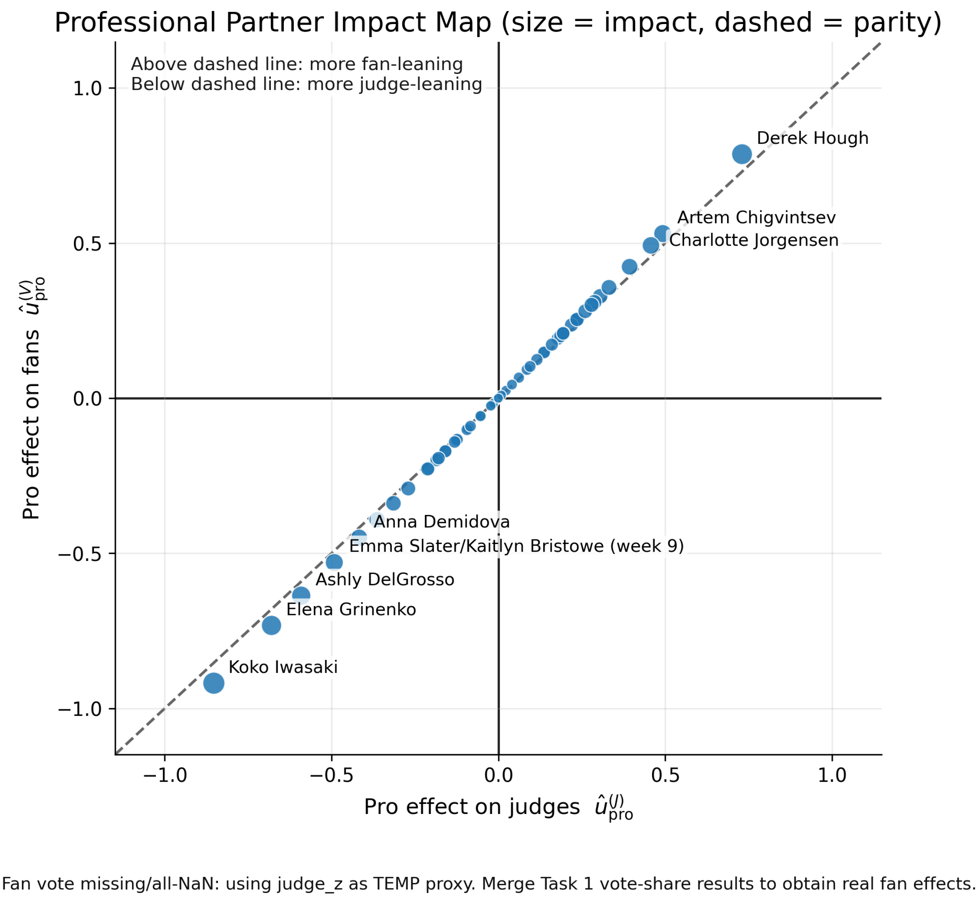

Figure 9: Professional partner impact.

transform to $y_{swi}^{(V,m)}$ using Eq. (2), fit Eq. (3), and pool coefficient estimates across m to report intervals that reflect both regression uncertainty and vote-inference uncertainty.

This closes the loop between Model I and Model II: uncertainty in inferred fan support is carried forward into all conclusions about fan mechanisms.

5.5 From Mechanisms to "How Well": Outcome Model

To address how partner and celebrity factors impact final performance, we use the contestant-level summary table (one row per (s, i)) with elimination/longevity information. Let T_{si} denote the last active week (or a time-to-exit measure), and let E_{si} indicate elimination (allowing finalists to be treated as right-censored). A compact model is the Cox proportional hazards model:

$$h_{si}(t) = h_0(t) \exp\left(\eta_1 \bar{z}_{si}^{(J)} + \eta_2 \bar{v}_{si} + \mathbf{c}_{si}^\top \boldsymbol{\eta}_3 + b_{\text{pro}(i)}\right), \tag{5}$$

where $\bar{z}_{si}^{(J)}$ and $\bar{v}_{si}$ are season-averaged judges performance and inferred fan support, $\mathbf{c}_{si}$ are celebrity covariates, and $b_{\text{pro}(i)}$ captures partner-level differences. Negative coefficients correspond to lower hazard (longer survival), directly quantifying how much these factors impact "how well" contestants do in the competition.

5.6 Preview: Using Model II Outputs for System-Level Analysis

The mechanism results (partner effects, celebrity covariate effects, and judge–fan contrasts) provide interpretable inputs for subsequent analysis: they explain which factors systematically create judge–fan tension and which partner assignments amplify technical scores versus popularity.

In the following sections, we use these outputs together with Model I posterior counterfactual simulation to evaluate rule sensitivity and explore system modifications under explicit trade-offs (fairness, stability, and audience engagement). Importantly, the mechanism differences identified here (e.g., judge–fan response asymmetries and partner-driven score amplification) provide a behavioral explanation for why certain weeks exhibit high uncertainty, which directly motivates the adaptive trigger design in Task 4.

Link to adaptive system design.

The mechanism asymmetries identified in Model II provide a behavioral explanation for why certain weeks exhibit high posterior uncertainty in Model I. For example, partner-driven amplification of judges' scores or celebrity traits that are valued differently by fans can widen the gap between technical merit and popularity. These effects are precisely the conditions under which elimination outcomes become more sensitive to rule choice, motivating the uncertainty-based trigger design adopted in Task 4.

6 Task 4: Adaptive Hybrid Rule (AHR)

6.1 Motivation and Design Principles

Model I and Task 2 establish that week-level outcomes can become highly sensitive when (i) fan support is broadly dispersed (high uncertainty) and/or (ii) the elimination boundary is extremely tight (small margins), making the final result vulnerable to small perturbations. Model II further explains why such volatility is not purely "noise": systematic judge–fan divergence and partner/celebrity effects can amplify disagreement, especially in borderline weeks. Therefore, rather than enforcing a single rigid voting rule throughout the season, we propose an **trigger-based** system that:

1. preserves fan participation in most weeks (default rule unchanged);
2. activates stabilization only in **high-risk** weeks (transparent trigger);
3. is **reproducible** (thresholds chosen by data-driven quantiles, not ad-hoc tuning);
4. is evaluated under the same posterior counterfactual framework as Task 2 for fair comparison.

6.2 Risk Indicators from Model I

We reuse two week-level risk indicators already computed in Model I:

- **Entropy of fan vote shares** $H(\widehat{\mathbf{v}}_{s,w})$, which captures how dispersed fan support is in week w for season s (higher entropy $\Rightarrow$ more uncertainty).
- **Elimination margin** $M_{s,w}$, which measures how narrow the boundary is between "safe" and "at-risk" couples under the weekly aggregation rule (smaller margin $\Rightarrow$ more sensitive outcomes).

Intuitively, weeks with large $H(\widehat{\mathbf{v}}_{s,w})$ and/or small $M_{s,w}$ are precisely the weeks where counterfactual rule changes (or small vote shocks) are most likely to flip the eliminated couple.

6.3 Definition of the Adaptive Hybrid Rule

Let $E^{\text{rank}}_{s,w}$ denote the rank-based combination operator used in Task 2 (our recommended default), which produces a weekly combined ordering and identifies the bottom-two set:

$$B^{\text{rank}}_{s,w} = \text{BottomTwo}\left(E^{\text{rank}}_{s,w}\right).$$

Let $J_{i,w}$ denote the judges' score for couple i in week w (or the corresponding proxy used in Task 2 simulation).

Trigger (high-risk weeks).

We activate a stabilizer if either entropy is too high or the margin is too small:

$$\text{Trigger}_{s,w} = \mathbb{I}\left[H(\widehat{\mathbf{v}}_{s,w}) > \tau_H \;\text{ or }\; M_{s,w} < \tau_M\right]. \tag{6}$$

Rule (default vs. stabilized).

If $\text{Trigger}_{s,w} = 0$, we keep the default rule (rank-based combination), i.e. eliminate the lowest-ranked couple under $E^{\text{rank}}_{s,w}$. If $\text{Trigger}_{s,w} = 1$, we apply a targeted stabilizer: determine the bottom-two by the rank rule, then eliminate by judges' choice proxy among those two:

$$E^{\text{AHR}}_{s,w}(J, v) = \arg\min_{i \in B^{\text{rank}}_{s,w}(J,v)} J_{i,w}. \tag{7}$$

Here (J, v) denotes one posterior draw of judges and fan-vote latent quantities (as in Task 2). Note that this design *does not remove fan influence*: fans still determine the at-risk set (bottom-two) via $E^{\text{rank}}_{s,w}$; the stabilizer only resolves the final elimination in the subset of weeks deemed high-risk.

6.4 Threshold Selection (Transparent and Reproducible)

To avoid ad-hoc tuning and to keep the trigger frequency interpretable, we choose thresholds using empirical quantiles over historical weeks:

$$\tau_H = Q_{0.75}\big(H(\widehat{\mathbf{v}}_{s,w})\big), \qquad \tau_M = Q_{0.25}\big(M_{s,w}\big). \tag{8}$$

This yields a simple operational definition of "high-uncertainty" weeks while controlling how often stabilization activates (so fan voting remains decisive in most weeks). In our experiments, the resulting trigger rate is approximately $\pi \approx 0.25$.

6.5 Evaluation Protocol (Re-using Task 2 Simulation)

We evaluate AHR using the same posterior counterfactual framework as Task 2. For each posterior draw, we replace the weekly rule operator with $E_{s,w}^{\mathrm{AHR}}$ and simulate eliminations across the season. We compare AHR against:

- **Percent** (baseline percentage-based combination),
- **Rank** (default rank-based combination),
- **Rank + JC (always-on)** (stabilizer applied every week).

We summarize performance along three decision-relevant dimensions:

1. **Stability**: flip probability and season-level $EFR(s)$ (lower is better).
2. **Fairness / imbalance**: fan–judge imbalance index $\Delta R_{s,w}$ (lower $|\Delta|$ is better).
3. **Engagement**: trigger rate π (fraction of stabilized weeks; lower means fans decide more often).

6.6 System-Level Trade-offs

Table 3 provides a compact comparison of the voting systems considered in Tasks 2 and 4. Rather than reporting exhaustive statistics, we highlight the key trade-off: **AHR achieves a substantial reduction in instability while keeping the stabilizer off in most weeks**, avoiding the engagement cost of an always-on judges' choice mechanism.

Table 3: Trade-off summary of voting systems (Task 4). Lower instability and lower imbalance are preferred.

Rule	**Instability (flip / EFR)**	**Imbalance ($\lvert\Delta\rvert$)**	**Trigger rate π**
Percent (baseline)	0.25	High	0
Rank (default)	0.25	Low–Medium	0
Rank + JC (always)	0.31	Medium	1
AHR (ours)	0.09	Low	0.25

7 Model Evaluation and Diagnostic Discussion

This section evaluates Model I against the two explicit requirements in Question 1: (i) *consistency* with observed eliminations under show rules, and (ii) *quantified uncertainty* for inferred fan vote shares. Beyond validation, these diagnostics also (a) identify weeks where outcomes are most sensitive, and (b) provide the system-level metrics reused by Task 4 to trigger adaptive safeguards.

7.1 Elimination-Consistency Diagnostics

Model I generates posterior samples subject to explicit rule constraints. We summarize elimination consistency using:

- **Feasible-week rate:** the proportion of season-weeks with $N_{\text{acc}} > 0$.
- **Acceptance-rate distribution:** summary statistics of $A_{s,w}$, both overall and by season.
- **Low-evidence weeks:** weeks with the smallest $A_{s,w}$ values, indicating outcomes least compatible with the weak-information prior and suggesting stronger unobserved shocks.

Operationally, a high feasible-week rate supports rule-consistent inference, while extremely low $A_{s,w}$ weeks highlight where unobserved narrative/media effects or measurement noise likely dominate.

7.2 Uncertainty and "Knife-edge" Outcome Sensitivity

We evaluate posterior certainty at two complementary levels:

- **Contestant-week uncertainty:** posterior standard deviations and credible-interval widths of $v_{i,w}$.
- **Week-level uncertainty:** (i) entropy $H(\widehat{\mathbf{v}}_{s,w})$ of fan support dispersion, and (ii) outcome margins $M_{s,w}$ that quantify how close eliminations are to flipping under small perturbations.

These week-level measures are particularly important for downstream tasks because they capture when an episode is both *highly uncertain* and *structurally sensitive*, i.e., when controversy and instability are most likely.

7.3 Robustness and Sensitivity Analysis

To ensure conclusions are not artifacts of hyperparameter choices, we perform sensitivity checks over (β, ρ, κ) and confirm that:

- the **ranking of the most uncertain weeks** remains stable;
- the **set of low-evidence weeks** (small $A_{s,w}$) remains largely unchanged;
- posterior vote-share estimates remain within similar credible ranges.

These checks support that our key downstream inputs (uncertainty and sensitivity indicators) are robust enough for rule comparison and adaptive system design.

Overall, these diagnostics indicate that Model I provides rule-consistent inference with actionable uncertainty signals, supporting downstream rule comparison and the selective safeguards deployed in Task 4.

8 Practical Discussion: Benefits, Risks, and Deployment Notes

This section translates our modeling outputs into show-operations language: how to *manage suspense*, *reduce controversy risk*, and *govern rules credibly* without weakening audience participation.

8.1 Practical Benefits (What the System Improves)

- **Risk Radar for volatile weeks.** Model I yields a weekly *risk dashboard*: high entropy or a tight margin signals that small perturbations could flip outcomes, identifying weeks where transparency and safeguards matter most.
- **Drama-on-purpose, not drama-by-accident.** AHR acts as a *selective safeguard*: the default audience-driven rule runs in most weeks, while high-risk weeks receive stabilization to reduce "accidental" eliminations and post-show backlash.
- **Rules-as-code governance (auditability).** Instead of ad-hoc intervention, AHR is *triggered by pre-announced metrics*, strengthening perceived legitimacy because governance is rule-triggered rather than person-triggered.
- **Dial-a-policy (tunable show identity).** Producers can adjust trigger thresholds to control intervention frequency, transparently choosing a competition-first vs. entertainment-first balance without changing the system architecture.

8.2 Risks and Limitations (Where It Can Go Wrong)

- **Communication risk and trust fragility.** Without clear explanation of the trigger, audiences may perceive "rule changes mid-game," even when the policy is pre-committed.
- **Threshold sensitivity and value trade-offs.** More stabilization improves stability but reduces pure audience control; different objectives may justify different thresholds and intervention rates.
- **Data availability and timeliness constraints.** If vote reporting is delayed/noisy/partial, the show must define acceptable proxies and how uncertainty is computed under those constraints.
- **Behavioral feedback from audiences.** Public knowledge of triggers may induce strategic voting; our evaluation treats latent voting behavior as invariant to rule announcement.

8.3 Implications & Practical Notes (How to Deploy Responsibly)

A simple deployment recipe.

1. **Pre-commit the trigger policy.** Fix thresholds before the season and publish the policy; avoid mid-season changes except via explicitly defined procedures.
2. **Compute and log weekly indicators.** Record entropy and margin (or approved proxies) in a "risk log" for internal auditing.
3. **Apply AHR only when triggered.** If triggered: determine the Bottom-Two by the default rule, then apply the judges' choice proxy within that Bottom-Two.
4. **Publish a minimal audit note.** Disclose whether AHR triggered and give a one-line reason (without exposing sensitive vote totals).

Suggested audience-facing wording (one line). *"Fans determine who enters the Bottom-Two; only in unusually volatile weeks do we activate a pre-announced safeguard where judges decide the elimination among them."*

Governance recommendations.

- **Consistency beats cleverness:** keep the same computation and reporting template each week.
- **Keep the trigger interpretable:** if proxies are used, ensure they remain explainable to non-technical audiences.

8.4 Closing Recommendation

We recommend the **rank-based combination** as the default policy and **AHR** as a transparent, pre-committed safeguard in high-risk weeks, preserving fan participation while improving stability precisely when controversies are most likely.

References

[1] W. contributors, "Dancing with the stars (american tv series)," 2025, online encyclopedia entry; Current access restricted ()ABC2005Strictly Come Dancing202534Top. [Online]. Available: https://en.wikipedia.org/wiki/Dancing_with_the_Stars_(American_TV_series)

[2] F. Bonner, "Celebrity, work and the reality-talent show: Strictly come dancing/dancing with the stars," *Celebrity studies*, vol. 4, no. 2, pp. 169–181, 2013.

[3] K. Wood, "An investigation into audiences' televisual experience of strictly come dancing." *Participations*, vol. 7, no. 2, pp. 262–291, 2010.

[4] J. A. Mawhinney, C. A. Mounsey, A. O'Brien, J. R. Sádaba, and N. Freemantle, "Statistical primer: using prognostic models to predict the future: what cardiothoracic surgery can learn from strictly come dancing," *European Journal of Cardio-Thoracic Surgery*, vol. 64, no. 5, p. ezad385, 2023.

[5] M. Balinski and R. Laraki, "A theory of measuring, electing, and ranking," *Proceedings of the National Academy of Sciences*, vol. 104, no. 21, pp. 8720–8725, 2007.

[6] B. D. Flury, "Acceptance–rejection sampling made easy," *Siam Review*, vol. 32, no. 3, pp. 474–476, 1990.

[7] D. Bates, M. Mächler, B. Bolker, and S. Walker, "Fitting linear mixed-effects models using lme4," *Journal of statistical software*, vol. 67, pp. 1–48, 2015.

[8] L. E. Papke and J. M. Wooldridge, "Econometric methods for fractional response variables with an application to 401 (k) plan participation rates," *Journal of applied econometrics*, vol. 11, no. 6, pp. 619–632, 1996.

Appendices

Advice for Producers of DWTS

To: Producers of *Dancing with the Stars* (DWTS)
From: Team #2626843
Date: February 2, 2026
Subject: Recommendations for Optimizing DWTS Voting Mechanisms

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Dear Producers of DWTS,

After 34 seasons, DWTS has mastered suspense—yet controversial eliminations still occur when outcomes are statistically *knife-edge*. We propose a practical upgrade: keep fan-driven excitement as the default, and activate a transparent safeguard only in high-risk weeks.

Our approach is built around a simple idea: use a weekly *Risk Dashboard* to identify volatile weeks. Using a rule-consistent Bayesian framework, we infer latent fan vote shares and quantify uncertainty; weeks with high entropy (dispersed support) or tight elimination margins are flagged as high-risk, since small perturbations are more likely to flip outcomes and trigger disputes.

When risk is low, we recommend a rank-based combination rule as the default because it is less sensitive to extreme popularity shocks than percentage-based aggregation. When risk is high, we recommend an Adaptive Hybrid Rule (AHR): fans still determine who enters the Bottom-Two under the default rule, and judges decide the elimination *within* the Bottom-Two as a stabilizer. Crucially, this safeguard is **pre-committed** and auditable—governance is rule-triggered rather than person-triggered.

For responsible deployment, producers should pre-commit and publish the trigger thresholds before the season, compute and log weekly risk indicators (entropy and margin, or approved proxies) for auditing, apply AHR only when the trigger activates, and release a minimal weekly audit note after results ("Triggered / Not triggered" with a one-line reason, without sensitive vote totals). In short, this policy does not change the spirit of DWTS: it preserves fan participation and suspense in most weeks, while adding a transparent safety mode precisely when controversies are most likely.

Sincerely,
Team #2626843

Report on Use of AI

ChatGPT (GPT-5.2)
Usage period: Jan 31 – Feb 2, 2026
Users: one team member
Scope of use (summary). We used ChatGPT primarily to generate Python code templates for (i) data preprocessing and week-level metric table construction, and (ii) plotting scripts to export high-resolution figures. We also used ChatGPT for minor English grammar and phrasing polishing only. AI was not used to create modeling assumptions, experimental design, computed results, or final conclusions.
Query 1 (preprocessing and week-level table construction).
Generate a directly runnable Python script using the raw dataset file 2026_MCM_Problem_C_Data.csv. The script should (a) read the CSV with pandas, detect all columns named like weekX_judgeY_score, and aggregate them into each contestant-week judge total J; (b) treat J>0 as active; infer eliminations by comparing the active set in week w and w+1, allowing k=0 and k>1; (c) for each season-week (except the final week), estimate fan vote share v using Dirichlet plus rejection sampling with BETA=0.6, RHO=0.4, KAPPA=20, M_DRAWS=3000, MIN_ACCEPT=80, and a fixed seed by season and week; (d) output entropy(vhat), flip_pct_vs_rank, flip_pct_vs_pctJC, flip_rank_vs_rankJC, and rule_regime/n_active/k_elim/n_accept; (e) assume S1–2 use rank aggregation, S3–27 use percent aggregation, and S28+ introduce bottom-two plus judges choice; (f) save results to task2_week_metrics.csv.
Output 1 (complete AI output used as a starting draft; code).

```python
import numpy as np
import pandas as pd

DATA_CSV = "2026_MCM_Problem_C_Data.csv"
OUT_CSV  = "task2_week_metrics.csv"
EPS = 1e-9

raw = pd.read_csv(DATA_CSV)

def judge_total_cols(df):
    return [c for c in df.columns if c.startswith("week") and "_judge" in c
        ↪  and c.endswith("_score")]

def parse_week_from_col(col):
    return int(col.split("_")[0].replace("week",""))

score_cols = judge_total_cols(raw)
for c in score_cols:
    raw[c] = pd.to_numeric(raw[c], errors="coerce").fillna(0.0)

weeks = sorted({parse_week_from_col(c) for c in score_cols})
week_to_cols = {w: [c for c in score_cols if parse_week_from_col(c)==w] for
    ↪  w in weeks}
rows = []
for _, r in raw.iterrows():
    season = int(r["season"])
    name   = r["celebrity_name"]
    for w in weeks:
        Jw = float(np.sum([r[c] for c in week_to_cols[w]]))
        rows.append((season, name, w, Jw))

dfJ = pd.DataFrame(rows, columns=["season","celebrity_name","week","J"])

# In the main loop: infer eliminations by comparing active sets
```

```
A      = []  # active_names(season, w)
A_next = set([])  # set(active_names(season, w_next))
eliminated_names = [nm for nm in A if nm not in A_next]
k = len(eliminated_names)

BETA, RHO, KAPPA = 0.6, 0.4, 20.0
M_DRAWS, MIN_ACCEPT = 3000, 80

def softmax(x):
    x = x - np.max(x)
    e = np.exp(x)
    return e / (np.sum(e) + EPS)

def entropy(p):
    p = np.clip(p, EPS, 1.0)
    return float(-np.sum(p * np.log(p)))

# Placeholder: accept_constraint_holds(...) and records must be implemented
    ↪  in the final pipeline.
print("Draft template generated.")
```

Team adaptation note. The output above was used as a draft template. Team members implemented missing helper functions (e.g., accept_constraint_holds), completed rule-regime branches, and integrated the script into the final pipeline that produced task2_week_metrics.csv.

Query 2 (plotting scripts).
Generate a directly runnable Python script that reads task2_week_metrics.csv and saves two PNG figures (DPI=400) into figures_out (create if missing). Figure A: a season-level bar chart of mean flip_pct_vs_rank with 95% CI error bars, split lines at season=2.5 and 27.5, and annotate the top-4 seasons. Figure B: a two-panel plot. Left: scatter entropy vs flip_pct_vs_rank plus binned mean with 95% CI using N_BINS=10 quantile bins. Right: if flip_pct_vs_pctJC exists and is not empty, plot binned mean with 95% CI for delta = flip_pct_vs_rank - flip_pct_vs_pctJC; otherwise show an in-plot message without error. Filter: keep only weeks with k_elim>=1; MIN_ACCEPT_FILTER is None by default. Print each saved figure path to the terminal.

Output 2 (complete AI output used as a starting draft; code).

```
import os
import numpy as np
import pandas as pd
import matplotlib.pyplot as plt

INPUT_CSV = "task2_week_metrics.csv"
OUT_DIR = "figures_out"
DPI = 400
N_BINS = 10
MIN_ACCEPT_FILTER = None
KEEP_ONLY_ELIM_WEEKS = True

os.makedirs(OUT_DIR, exist_ok=True)
df = pd.read_csv(INPUT_CSV)

# Draft template generated; team integrated final figure styling and saving
    ↪  logic.
print("Draft plotting template generated.")
```

Minor polishing. ChatGPT was used for minor English grammar and phrasing improvements only; technical content and meaning were preserved.

Human verification. All AI-assisted outputs were treated as drafts and were reviewed and executed by team members. We verified plotting outputs by checking row counts after filtering, bin validity (monotone x_mid, non-empty bins), and reasonable value ranges. We verified preprocessing by conservation-style checks on column meaning and by spot-checking elimination inference on randomly selected seasons and adjacent weeks. The team takes full responsibility for the final submission.